Alan Siby

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Education

University of Texas at Austin Aug 2026 - May 2030
Bachelor of Science in Mechanical Engineering *Austin, TX*

Round Rock High School & Austin Community College (Dual Enrollment) Aug 2022 – May 2026
Coursework: Calculus, Physics, and other AP/Advanced courses *Round Rock*

Research Experience

Research Intern, Institute for Computing in Research (ICR) Jun 2025 – Jul 2025

- Designed and implemented a Python-based simulation of a 1-axis spacecraft attitude control system, modeling reaction wheel dynamics, actuator saturation, and proportional-derivative (PD) control
- Built a grid-search optimization routine to tune controller gains against settling time, overshoot, and control torque
- Diagnosed and corrected flawed performance-metric logic by systematically verifying simulation output against expected physical behavior, rather than trusting results that appeared numerically clean
- Co-authored findings and published on engrXiv, contributing to open-source aerospace controls research
- Collaborated with mentors and peers to refine algorithms and present results in professional technical format

Projects

1-Axis Spacecraft Attitude Control: PD Design with Reaction Wheel Saturation ICR, 2025

- Built a fourth-order Runge-Kutta simulation framework in NumPy/Matplotlib to model closed-loop spacecraft dynamics under real actuator constraints
- Verified controller performance using root locus and time-domain analysis across six spacecraft configurations
- Identified and resolved a critical flaw in settling-time detection logic that produced false-positive stability results

Magic Snap: Ergonomic & Magnetic-Charging Shell for the Apple Magic Mouse Team Flow State, 2026

- Co-designed and CAD-modeled a 3D-printed ergonomic shell addressing RSI risk and a magnetic charging dock resolving the device's original bottom-mounted charging flaw
- Directed user research and a weighted decision matrix across three competing team concepts to guide design direction
- Iterated through multiple prototypes, refining palm-arch fit, mouse glide, and charging-cable management based on user testing and interviews

Puzzle Cube Intro to Engineering & Design, 2022

- Designed a 3D wooden puzzle system in Autodesk Inventor, engineering interlocking geometry from scrap hardwood cubes
- Iterated part tolerances and joint geometry through prototyping to ensure a precise, snug fit across all puzzle pieces

Experience & Activities

Store Associate, The Home Depot Mar 2026 – Aug 2026

- Help customers find plumbing products and answer questions about pipes, fittings, and fixtures
- Maintain a clean, organized, and safe work area while supporting team operations and customer service

Founder & Basketball Trainer, Apex Basketball Training Sep 2024 – May 2026

- Founded and independently operate a youth basketball training business coaching 50+ athletes; manage scheduling, client communication, and marketing
- Hosted over 75 sessions with various groups and levels

NASA High School Aerospace Scholar Oct 2024 – Jun 2025

- Selected for NASA's competitive national Aerospace Scholars program; collaborated on a team designing satellite enhancements for lunar exploration

Awards

National Merit Commended (2025) | AP Scholar with Distinction (2025) | National Honor Society (2025) | AP Scholar (2024) | DECA SCDC State Qualifier (2025)

Skills

Programming: Python, Java, Matplotlib, NumPy
Tools: Autodesk Inventor (CAD), Fusion 360, LaTeX
Research: Technical writing, data analysis, control systems modeling